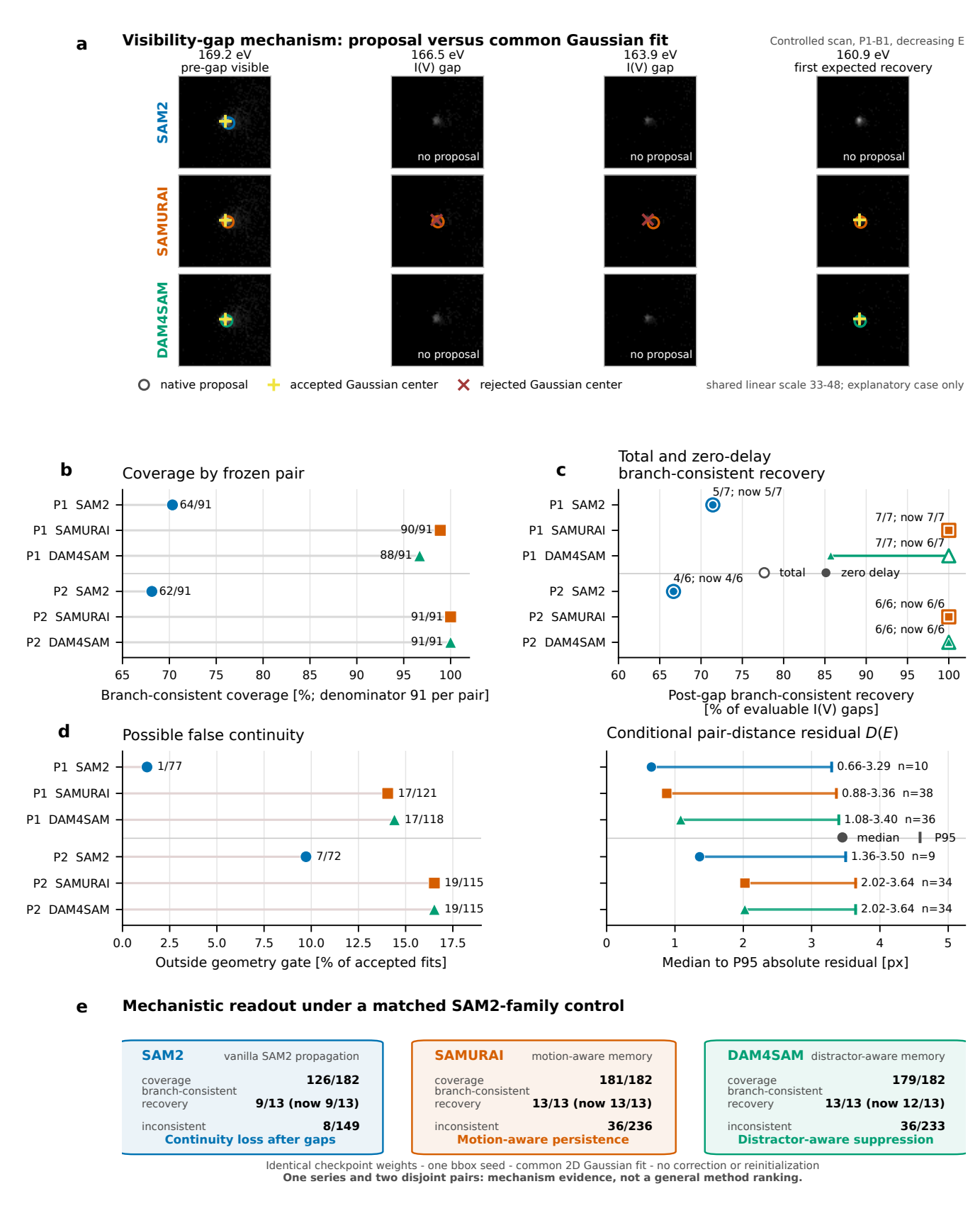
a
Visibility-gap mechanism: proposal versus common Gaussian fit
Controlled scan, P1-B1, decreasing E
169.2 eV
pre-gap visible
166.5 eV
I(V) gap
163.9 eV
I(V) gap
160.9 eV
first expected recovery
SAM2
SAMURAI
DAM4SAM
no proposal
native proposal
accepted Gaussian center
rejected Gaussian center
shared linear scale 33-48; explanatory case only
b
Coverage by frozen pair
P1 SAM2
P1 SAMURAI
P1 DAM4SAM
P2 SAM2
P2 SAMURAI
P2 DAM4SAM
64/91
90/91
88/91
62/91
91/91
91/91
Branch-consistent coverage [%; denominator 91 per pair]
c
Total and zero-delay
branch-consistent recovery
5/7; now 5/7
7/7; now 7/7
7/7; now 6/7
4/6; now 4/6
total
zero delay
6/6; now 6/6
6/6; now 6/6
Post-gap branch-consistent recovery
[% of evaluable I(V) gaps]
d
Possible false continuity
1/77
17/121
17/118
7/72
19/115
19/115
Outside geometry gate [% of accepted fits]
Conditional pair-distance residual D(E)
0.66-3.29 n=10
0.88-3.36 n=38
1.08-3.40 n=36
median
P95
1.36-3.50 n=9
2.02-3.64 n=34
2.02-3.64 n=34
Median to P95 absolute residual [px]
e
Mechanistic readout under a matched SAM2-family control
SAM2
vanilla SAM2 propagation
coverage 126/182
branch-consistent recovery 9/13 (now 9/13)
inconsistent 8/149
Continuity loss after gaps
SAMURAI
motion-aware memory
coverage 181/182
branch-consistent recovery 13/13 (now 13/13)
inconsistent 36/236
Motion-aware persistence
DAM4SAM
distractor-aware memory
coverage 179/182
branch-consistent recovery 13/13 (now 12/13)
inconsistent 36/233
Distractor-aware suppression
Identical checkpoint weights - one bbox seed - common 2D Gaussian fit - no correction or reinitialization
One series and two disjoint pairs: mechanism evidence, not a general method ranking.